

Shishir Somir Mukherjee

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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate with proficiency in **C, C++, Python, Java**, and **SQL**. Experienced in **machine learning, data analysis**. Skilled at combining **low-level programming** with **data-focused** approaches to create efficient and reliable solutions.

EDUCATION

Vellore Institute Of Technology

Bachelor of Technology in Computer Science and Engineering

Kendriya Vidyalaya No.1 Shahibaug

XII (Senior Secondary)

Kendriya Vidyalaya No.1 Shahibaug

X (Secondary)

Bhopal, Madhya Pradesh

Sep 2022 – Ongoing

Ahmedabad, Gujarat

June 2021 – July 2022

Ahmedabad, Gujarat

June 2019 – May 2020

PROJECTS

Smart Inventory Routing Dashboard | Streamlit, Random Forest, APIs

July 2025 – July 2025

- Developed a real-time dashboard to optimize inter-store inventory transfers across Walmart locations.
- Forecasted 3-day demand per store by training a Random Forest model on 500+ synthetic sales records, factoring in weather and traffic.
- Incorporated Google Maps, and VisualCrossing APIs to account for product availability, live traffic, and weather-driven demand shifts.
- Reduced stockouts by 15% and improved on-shelf availability by 20% using urgency-based filtering and transfer approvals.

Skin Cancer Detection Model | TensorFlow, CNN, Python

Nov 2024 – April 2025

- Designed a CNN model to classify 2,000+ dermoscopic images into 7 skin lesion categories, achieving 79% test accuracy.
- Enhanced performance using image preprocessing and augmentation (resizing, rotation, width/height shift, shear, zoom, horizontal flip), increasing generalization by 18%.
- Constructed a custom CNN with 3 convolutional layers, batch normalization, max pooling, and dense layers for robust feature extraction.
- Applied model checkpointing, saving the best model after 20 epochs, enabling robust early detection of skin cancer.

Diabetes Prediction Model | Python, KNN, XGBoost, SVM

April 2024 – June 2024

- Engineered ML models (KNN, XGBoost, SVM) on a dataset of 7,000+ patient records, achieving 91% accuracy in predicting diabetes onset.
- Cleaned and transformed data through missing value imputation, normalization, feature scaling, and class balancing, handling 5% missing data.
- Evaluated models using accuracy, precision, recall, and ROC-AUC, identifying XGBoost as the top-performing classifier.
- Optimized 15 key features iteratively, reducing false positives by 12% and improving reliability for clinical decision-making.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, OOP, SQL, HTML, CSS, JS

Development Tools: Github, Jupyter Notebook, VS Code, Visual Studio, PyCharm

Libraries: Pandas, NumPy, Matplotlib, Sci-kit Learn, Tensorflow, STL, API

ACHIEVEMENTS

- Won 1st place in a college-level Hackathon organized by GeeksforGeeks VIT Bhopal, competing among 50+ teams.